1. A KNN classifier has excellent performance on the training data but noticeably worse performance on new data. Which change would be most likely to help?
    a. Decrease *k*
    **b. Increase *k***
    c. Give variables with larger variances more weight
    d. Include an interaction term

**Solution: A KNN model that is overfitting is likely because the value of *k* is small. Increasing *k* will likely improve performance. Decreasing *k* will only make the problem worse. Weighting variables with larger variances more heavily will worsen performance. Including an interaction term is not necessary (KNN is nonparametric).**

2. A student says, "The most flexible model is always preferable because it can represent more complicated relationships." Explain what is wrong with this reasoning.

**Solution: This is not true. There are many cases where the most flexible model is not preferable, either because the more flexible model is less interpretable and the goal is interpretation, or because the more flexible model can fit too closely to patterns in the training data that do not exist in the population, leading to overfitting (low training error and high testing error).**

3. You fit Models A, B, and C:

| Model | Training accuracy | Testing accuracy |
|---|---|---|
| **A** | 71% | 70% |
| **B** | 86% | 82% |
| **C** | 100% | 68% |

Which model would you prefer based only on this information? What appears to be happening with Model C?

**Solution: You should prefer Model B; it produces the highest testing accuracy. Model C appears to be overfitting. (Model B shows some overfitting also, but less so than Model C.)**

4. A student makes these three claims:
    a. A predictor with zero Pearson correlation with the outcome cannot improve prediction.
    b. Two highly correlated predictors necessarily interact.
    c. A model with lower training error is always preferable.

    For each claim, state whether it is true or false and explain.

**Solution: For (a), this is false. Predictors with zero Pearson correlation with the outcome do not have a linear correlation with the outcome but may very well be able to improve predictions in nonlinear and/or empirically driven models. For (b), this is also false.**

**Interactions have nothing to do with whether predictors are correlated. An interaction means the relationship between X1 and Y differs depending on the value of X2. For (c), this is also false. Training error is not a good estimator of testing error. As training error goes to zero, testing error almost always increases.**

5. A new observation has the following seven nearest neighbors, already ordered from closest to farthest:

| Neighbor | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Class | A | B | B | A | A | B | B |

Predict the new observation's class using $k$ = 1, 3, 5, and 7.

**Solutions: Using $k$ = 1, we predict A; using $k$ = 3, we predict B; using $k$ = 5, we predict A; using $k$ = 7, we predict B.**

6. A student standardizes age but forgets to standardize annual income before fitting KNN. Explain why this is a problem even though both variables are numeric.

**Solution: This is a problem because KNN is a distance-based algorithm. Income most likely has a much larger variance than age, and therefore will have a much larger impact on distance calculations if it is not also standardized.**

7. A researcher wants to use KNN to predict whether or not a bank customer will default on a loan. They write the following code:

```
loan_recipe <-
  recipe(defaulted ~ ., data = loans) |>
  step_dummy(all_nominal_predictors(), one_hot = TRUE)

loan_model <-
  nearest_neighbor(neighbors = 5) |>
  set_engine("kknn") |>
  set_mode("classification")

loan_fit <-
  loan_model |>
  fit(defaulted ~ ., data = loans)
```

Their data contain the following variables:
application_id
age

income
employment_type
credit_score
default_date
defaulted

Identify at least four problems with this analysis. For each problem, explain why it matters and how you would correct it.

**Solution: Here are all problems with this analysis, along with why they matter and how to correct:**

- **"." syntax will include all variables, including application_id, which is an identifier, not an actual predictor, and default_date, which would be explicit target leakage. Remove those predictors.**
- **There is no step_normalize(), which is important for KNN because it is distance-based. Add that to the recipe.**
- **The recipe is not actually being used; loan_recipe is defined, but the model is fit directly to the raw data.**
- **Employment_type therefore hasn't actually been dummy encoded for the fitted model.**

**Note that one hot encoding is not a problem for KNN; KNN has no regression intercept creating the linear dependency problem.**

8. A scatterplot of X and Y shows a very strong U-shaped relationship. Their Pearson correlation is 0.03. Which statement is best?
    a. X and Y are essentially unrelated
    b. The scatterplot is likely wrong
    **c. There is little linear correlation, despite a strong nonlinear relationship**
    d. X will not be a useful variable for predicting Y
    e. Both C and D

**Solution: The predictors have a strong U-shaped (nonlinear) relationship, but little linear correlation. This does not mean X will not be useful for predicting Y; nor does it mean they are unrelated. It also does not mean the scatterplot is wrong.**

9. Which EDA finding is evidence of an **interaction**?
    a. X1 and X2 have a Pearson correlation of 0.91
    **b. The relationship between X1 and Y is positive for one level of X2 and negative for the other**
    c. X1 is strongly right-skewed and X2 is strongly left-skewed

d. X1 and Y have a Pearson correlation of 0.80
e. The relationship between X1 and Y is the same across both levels of X2

**Solution: B is the definition of an interaction. Pearson correlation does not imply the existence of an interaction. Neither does skew.**

For Questions 10–12, suppose a university wants to predict whether a student will graduate within four years, using information available **at the end of the student's first year**.

| Variable | Description |
|---|---|
| StudentID | Unique identifier |
| HS_GPA | High school GPA |
| FirstYearGPA | GPA at end of first year |
| Major | Major at end of first year |
| CreditsAttempted | First-year credits attempted |
| CreditsCompleted | First-year credits completed |
| Dorm | First-year residence hall name |
| GraduationDate | Date of graduation, if graduated |
| FinalGPA | GPA upon leaving the university |
| Graduated4Years | Outcome |

10. Which variables clearly should **not** be included as predictors in the model? Select all that apply.
    **a. StudentID**
    b. HS_GPA
    c. FirstYearGPA
    **d. GraduationDate**
    **e. FinalGPA**
    f. Major
    **g. Graduated4Years**
    h. CreditsAttempted
    i. CreditsCompleted

**Solution: A is an ID variable. D and E are both variables that would not be known at the end of the first year. G is the outcome.**

11. Which predictors would require dummy encoding before fitting a logistic regression? Select all that apply.
    a. HS_GPA
    b. FirstYearGPA
    **c. Major**
    d. CreditsCompleted
    **e. Dorm**

**Solution: These are categorical variables, which would require dummy encoding.**

12. A student creates a new variable, CompletionRate. CompletionRate is calculated as the ratio of CreditsCompleted to CreditsAttempted. Would including CompletionRate as a predictor constitute target leakage? Explain.

**Solution: No. CompletionRate is just the proportion of credits that a student attempted in their first year that they were actually able to complete, and it is not leakage (since the outcome is graduation at the end of four years). One could argue, however, that CompletionRate, CreditsCompleted, and CreditsAttempted should not all be included as predictors simultaneously (though that is not the question being asked).**

13. A student wants to predict "salary" using "experience" and "department." They fit the following model:

```
salary_recipe <- recipe(salary ~ experience + department,
                        data = employees) %>%
  step_dummy(department, one_hot = TRUE)

salary_model <- linear_reg() %>%
  set_engine("lm")

salary_workflow <- workflow() %>%
  add_recipe(salary_recipe) %>%
  add_model(salary_model)

salary_fit <- fit(salary_workflow, data = employees)
```

Suppose "department" has three levels, "Sales," "IT," and "HR."

After fitting the model, R reports that the model matrix is rank-deficient and that one or more coefficients cannot be uniquely identified.

a. What caused the problem?

**Solution: One-hot encoding; this creates three indicator variables, one per level of "department," which results in a linear dependency that OLS regression cannot handle.**

b. Why does "one_hot = TRUE" create this problem for an ordinary linear regression model with an intercept?

**Solution: Because the model contains an intercept, which is itself a column of ones. Therefore, the design matrix is not full rank.**

c. How should the code be changed?

**Solution: Set "one_hot = FALSE."**

d. After making this change, how many indicator variables will be created for "department"?

**Solution: Two.**

14. A logistic regression model predicts whether a patient has pneumonia based on symptoms and clinical measurements. For one patient, the model produces a predicted probability of 0.78. Which interpretation is correct?
    a. The model is 78% accurate for this patient.
    b. The patient definitely has pneumonia.
    **c. The fitted model assigns this patient a probability of 0.78 of belonging to the pneumonia class.**
    d. 78% of the patient's symptoms indicate pneumonia.

**Solution: The model cannot be "78% accurate" for an individual patient. Nor does this predicted probability tell us anything about the patient's symptoms. We cannot be absolutely certain the patient either does or doesn't have pneumonia. C is the correct definition.**

15. A model predicts whether borrowers will default on a loan. Four borrowers receive the following predicted probabilities:

| Borrower | Predicted Probability |
|---|---|
| A | .31 |
| B | .48 |
| C | .63 |
| D | .81 |

a. Using a threshold of 0.50, which borrowers would be classified as likely to default?

**Solution: Borrowers C and D.**

b. Using a threshold of 0.70, which borrowers would be classified as likely to default?

**Solution: Only Borrower D.**

c. Explain what happens to the number of positive classifications when the threshold is increased.

**Solution: When the threshold is increased, the number of positive classifications will tend to decrease.**

16. A logistic regression predicts whether a customer will cancel a subscription. The coefficient for the predictor MonthlyFee is 0.04.

    A student writes: "A $1 increase in the monthly fee causes a customer's probability of canceling to increase by 4%."

    Identify **one problem** with this interpretation and provide a better interpretation.

    **Solution: This statement implies that there is a linear relationship between monthly fee and probability of cancellation. There is not. There is a linear relationship between monthly fee and log odds of cancellation, or logit. Also, we cannot assume that we can draw causal conclusions. A better interpretation would be either "A $1 increase in the monthly fee is associated with a 0.04 increase in log odds," or "An increase in the monthly fee is associated with an increased probability of cancellation" (this is known because the sign of the coefficient is positive).**

17. A model produces the following confusion matrix:

| | **Actually Positive** | **Actually Negative** |
|---|---|---|
| **Predicted Positive** | 72 | 18 |
| **Predicted Negative** | 28 | 182 |

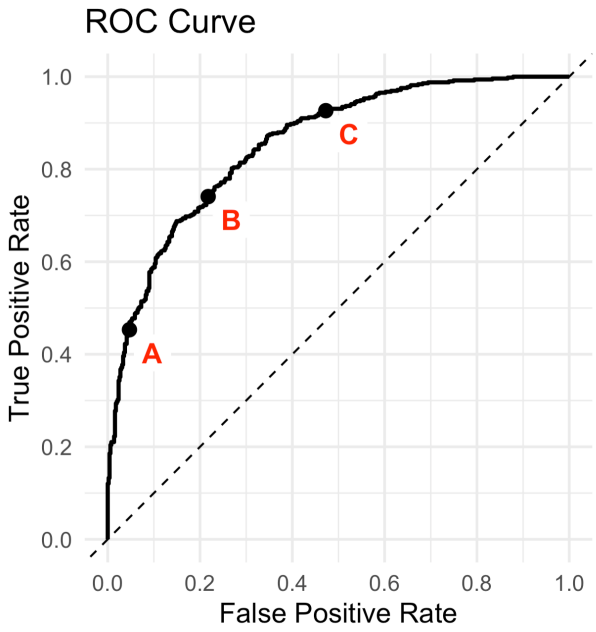
Calculate accuracy, true positive rate, and false positive rate.

**Solution: Accuracy is (72 + 182)/(72 + 182 + 18 + 28) = 254/300 = 84.67% (or 0.8467). The true positive rate is 72/(72 + 28) = 72/100 = 72% (or 0.72). The false positive rate is 18/(18+182) = 9% (or 0.09).**

18. A student says: "I changed my classification threshold from 0.50 to 0.30, so the ROC AUC of my fitted model will change." Is this correct? Explain.

**Solution: This is not correct. The ROC AUC measures the area under the ROC curve, which is a curve that plots true positive rates and false positive rates across thresholds. Therefore, using a different threshold value to classify observations will have no impact whatsoever on the area under the ROC curve.**

19. Suppose the ROC curve below represents a model used to screen patients for a very dangerous disease. A false negative is considered substantially worse than a false positive. Of points A, B, and C, which threshold would you be most inclined to choose?




a. Point A
b. Point B
**c. Point C**
d. All points are equally good
e. Not enough information to decide

**Solution: The problem implies that we are only interested in maximizing our true positive rate (and therefore, minimizing our false negative rate), regardless of our false positive rate. Therefore, we would prefer the threshold corresponding to Point C (despite a false positive rate of almost 50%).**

20. You have 50,000 observations, two predictors, and clear evidence that the covariance structures differ substantially across levels of your outcome variable. Which would you expect to have the greater potential to capture the true decision boundary?
    a. LDA
    **b. QDA**
    c. Both would do equally well
    d. Neither would do well because they are both parametric models

**Solution: LDA assumes that the covariance structure does not differ and therefore would not do well. QDA would likely excel despite estimating more predictors; *p* is not large, and there are a huge number of observations (so parametric models are fine).**